



University of Vermont

THE UNIVERSITY OF VERMONT
COLLEGE OF ENGINEERING AND MATHEMATICAL SCIENCES
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING

Technical Documentation Package Index

Team 485 - Autonomous Weeding Robot

ANDIE BROWN
NIELS KELLER
CADE LARCOM
JOHN POIRIER

Revision History

Date	Ver.	Description
04/29/2027	1.0	Initial Release

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Contents

1	Mechanical	1
1.1	System Level Assembly Drawing	1
1.2	System Level Bill of Materials	5
2	Electrical	7
2.1	System Level Wiring Diagram	7
2.2	Electrical BoM	7
2.3	Front Tool Board	8
2.3.1	PCB Schematic	8
2.3.2	PCB Drawings	9
2.3.3	PCB Bill of Materials	10
2.3.4	KiCad Files	11
3	Software Documentation	12
3.1	Front Tool	12
3.2	ROS2 Packages	12
3.3	Machine Learning	14
4	Design Files	16
5	User Manual	16

Mechanical

1.1 System Level Assembly Drawing

Files containing the overall mechanical assembly of the robot in several accessible formats.

PDF: Mechanical\AllMechanicalDrawings.pdf

DWG Files: Mechanical\DWGFiles\

STEP File: Mechanical\STEPFiles\

Water Jet Files: Mechanical\WaterJetFiles\

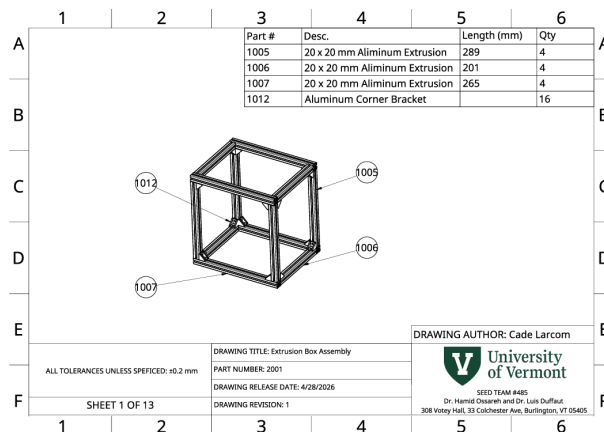


Figure 1. Extrusion Box Assembly Drawing

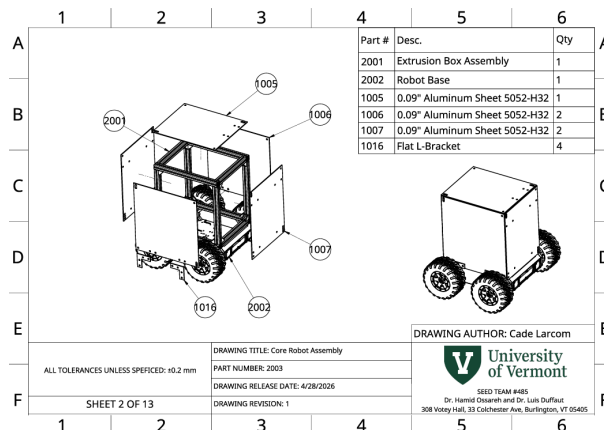


Figure 2. Core Robot Assembly Drawing

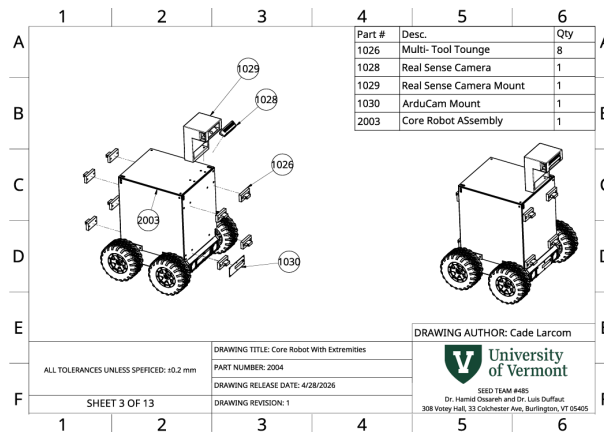


Figure 3. Core Robot with Extremities Drawing

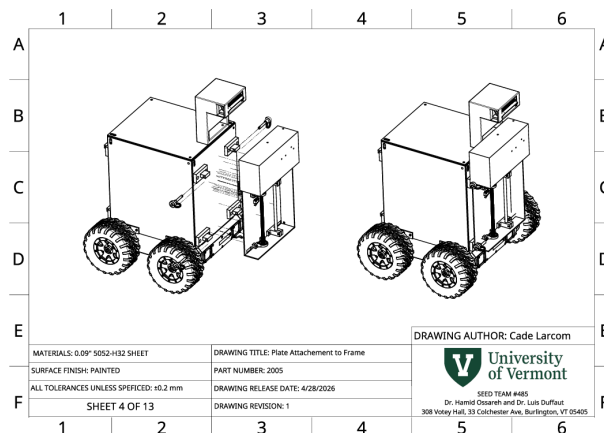


Figure 4. Plate Attachment to Frame Drawing

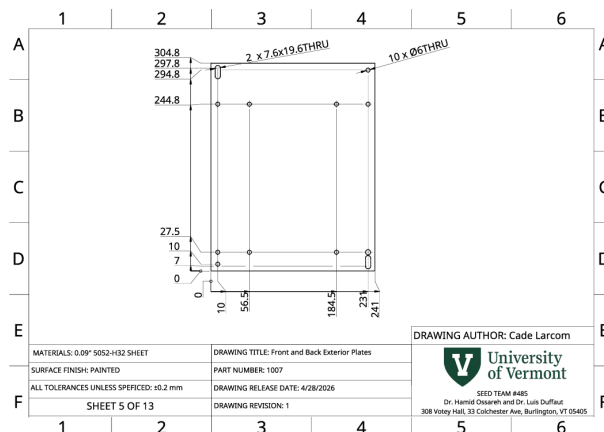


Figure 5. Front and Back Exterior Plates Drawing

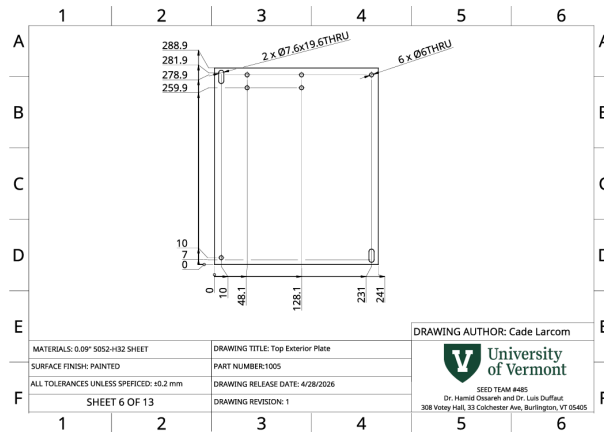


Figure 6. Top Exterior Plate Drawing

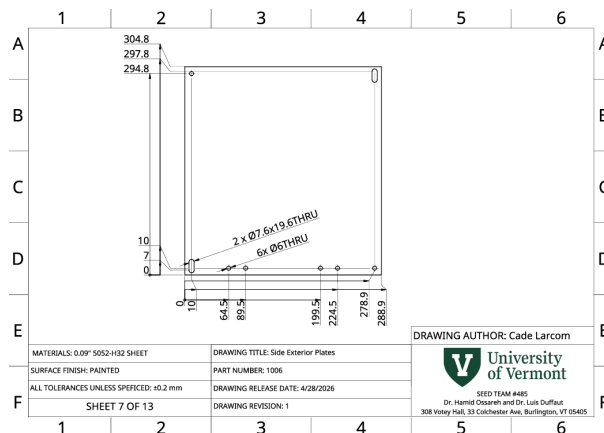


Figure 7. Side Exterior Plates Drawing

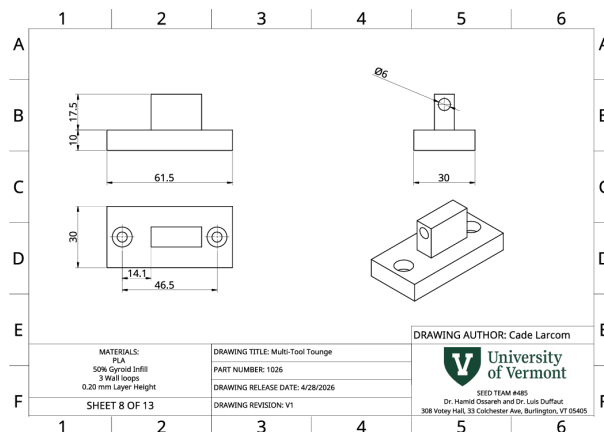


Figure 8. Multi-Tool Tongue Drawing

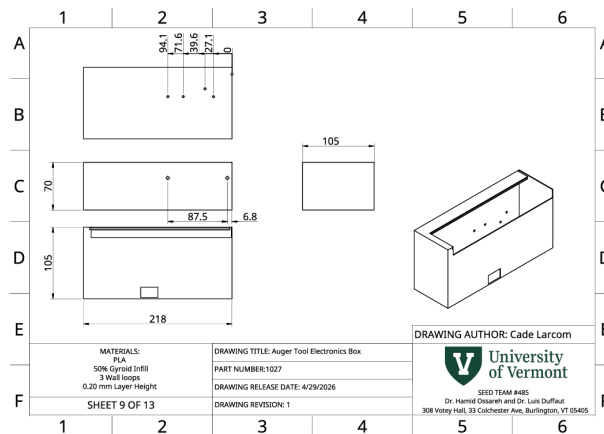


Figure 9. Auger Electronics Box Drawing

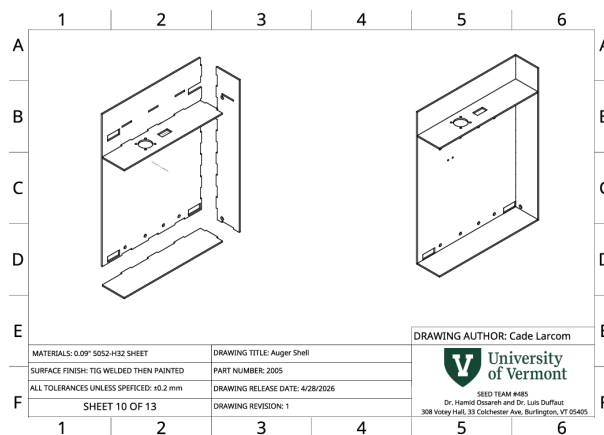


Figure 10. Auger Shell Drawing

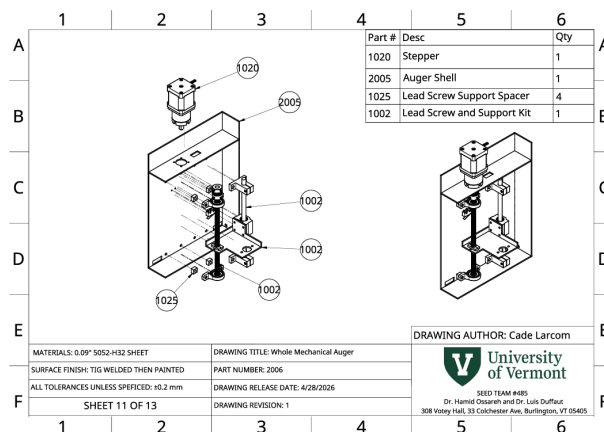


Figure 11. Whole Mechanical Auger Drawing

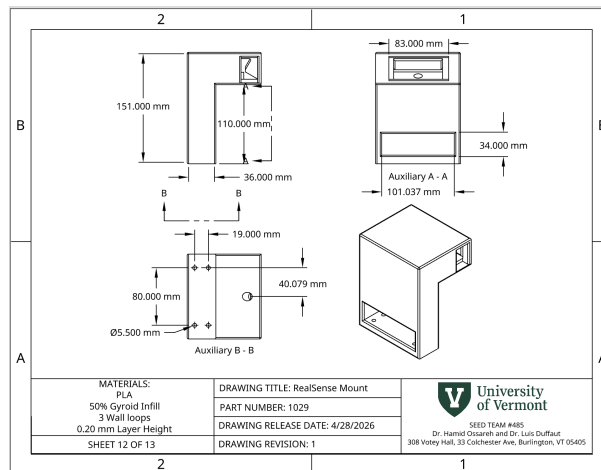


Figure 12. RealSense Camera Mount Drawing

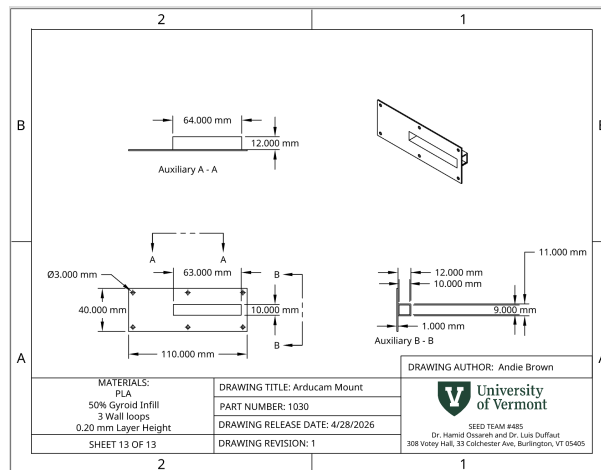


Figure 13. ArduCam Mount Drawing

1.2 System Level Bill of Materials

File containing the bill of materials for the full system level. The BoM is also shown in [Table 1](#).

Mechanical\Mechanical_BoM.xlsx

Table 1. Mechanical Bill of Materials

Item	PN	Qty	Description	MFG	Vendor	Cost (\$)
<i>Assembly 10 – Lead Screw and Support Kit</i>						
10	1002	1	Lead Screw and Support Kit	NUIBBYED	Amazon	29.99
10.1	1021	1	Lead Screw	NUIBBYED	Amazon	–
10.2	1022	1	Linear Beam	NUIBBYED	Amazon	–
10.3	1023	2	Lead Screw Support	NUIBBYED	Amazon	–
10.4	1024	2	Linear Beam Support	NUIBBYED	Amazon	–
10.5	1025	4	Lead Screw Support Spacer	NUIBBYED	Amazon	–
<i>Assembly 20 – Robot Base</i>						
20	2002	1	Robot Base (A4WD3-WH-KT)	Lynxmotion	RobotShop	575.95
<i>Assembly 30 – Extrusion Box</i>						
30.1	1009	4	11.375 in. Al Extrusion 20×20	McMaster-Carr	McMaster-Carr	26.08
30.2	1010	4	7.913 in. Al Extrusion 20×20	McMaster-Carr	McMaster-Carr	26.08
30.3	1011	4	10.425 in. Al Extrusion 20×20	McMaster-Carr	McMaster-Carr	26.08
30.4	1012	1	Al Corner Bracket Kit (20 set)	–	Amazon	14.99
<i>Assembly 40 – Core Robot Assembly</i>						
40.1	1005	1	0.09 in. Al Sheet (11.375×9.488 in.)	OnlineMetals	OnlineMetals	18.85
40.2	1006	2	0.09 in. Al Sheet (11.375×12 in.)	OnlineMetals	OnlineMetals	37.70
40.3	1007	2	0.09 in. Al Sheet (12×9.488 in.)	OnlineMetals	OnlineMetals	37.70
40.4	1016	4	Flat L-Bracket	RELIABILT	Lowes	15.92
40.5	1013	2	Krylon Green Spray Paint	Krylon	Lowes	13.26
40.6	1014	2	Krylon Yellow Spray Paint	Krylon	Lowes	13.26
40.7	1008	1	High Visibility Tape	WENFUMUSIC	Amazon	9.99
<i>Assembly 50 – Core Robot With Extremities</i>						
50.1	1026	8	Multi Tool Tongue (3D printed)	FabLab	FabLab	–
50.2	1003	4	Quick Release Pins (2-pack)	VXCICC	Amazon	59.96
50.3	1027	1	Electronics Holder (3D printed)	FabLab	FabLab	–
50.4	1028	1	RealSense Camera D435	Intel	Intel	314.00
50.5	1029	1	RealSense Camera Mount (3D)	FabLab	FabLab	–
50.6	1030	1	ArduCam Mount (3D printed)	FabLab	FabLab	–
50.7	1032	1	ArduCam	Arducam	Amazon	–
<i>Assembly 60 – Auger Shell</i>						
60	2005	1	Auger Shell	Machine Shop	–	–
<i>Assembly 70 – Whole Mechanical Auger</i>						
70.1	1031	1	Auger Bit	DEEGETEL	Amazon	19.99
70.2	1018	1	Time of Flight (VL53L1X)	SparkFun	SparkFun	29.95
70.3	1019	1	Limit Switch	LiebeWH	Amazon	11.71
70.4	1020	1	Stepper Motor (3327_0)	Phidgets	Phidgets	44.00
Total						1325.47

Items marked “–” for cost are included in the price of their parent kit assembly.

2 Electrical

2.1 System Level Wiring Diagram

This is the overall system wiring diagram for the main robot (Page 1) and front tool (Page 2). Electrical\Electrical_Wiring_Diagram.pdf

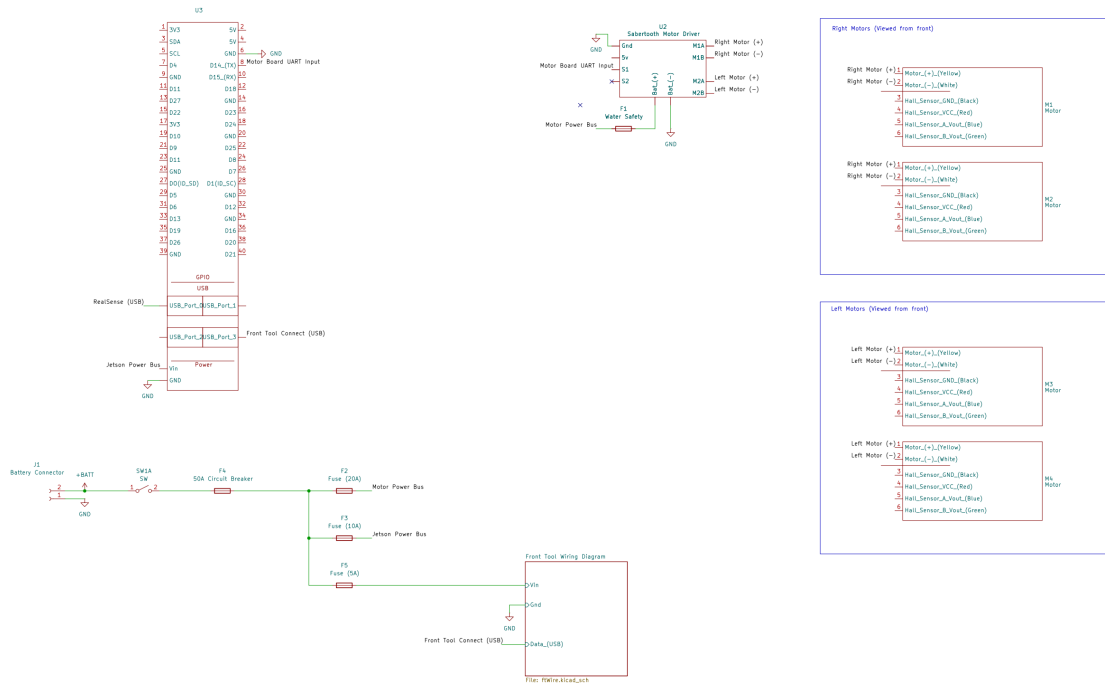


Figure 14. Electrical Wiring Diagram

2.2 Electrical BoM

File containing the bill of materials for the electrical system. The BoM is also shown in Table 2.

Electrical\System_Level_BOM.xlsx

Table 2. Electrical System Bill of Materials

Item	MFG PN	Qty	Description	MFG	Cost (\$)
<i>Compute & Control</i>					
1	22098	1	Yahboom Jetson Orin Nano Dev Kit	Yahboom / NVIDIA	369.00
2	ESP-WROOM-32	1	ESP32 Wi-Fi/BT Micro-controller	Espressif	10.00
<i>Motor Drive</i>					
3	A4WD3-WH-KT	1	Robot Base (motors incl.)	Lynxmotion	575.95
4	PN00218-DME2	1	Sabertooth 2×12 R/C Motor Driver	Dimension Eng.	80.00 [†]
5	IBT-2 (BTS7960)	1	H-Bridge Motor Driver, 43 A	HiLetgo / Generic	12.00 [†]
<i>Vision / Sensing</i>					
6	D435i	1	Intel RealSense Depth Camera	Intel	314.00
7	IMX179	1	Arducam 8MP Autofocus USB Camera	Arducam	35.00 [†]
<i>Power & Protection</i>					
8	CS-664A1	1	Fuse Holder	—	— [‡]
<i>Custom / Miscellaneous</i>					
9	Custom	1	Front Tool PCB	Custom	— [‡]
10	N/A	N/A	Cables and Electrical Connectors	Various	— ^{\$}
Total (priced items)					1,395.95

2.3 Front Tool Board

2.3.1 PCB Schematic

File containing the PCB schematics for the front tool board.

Electrical\FrontTool\Tool_Control_PCB_Rev0_Schm.pdf

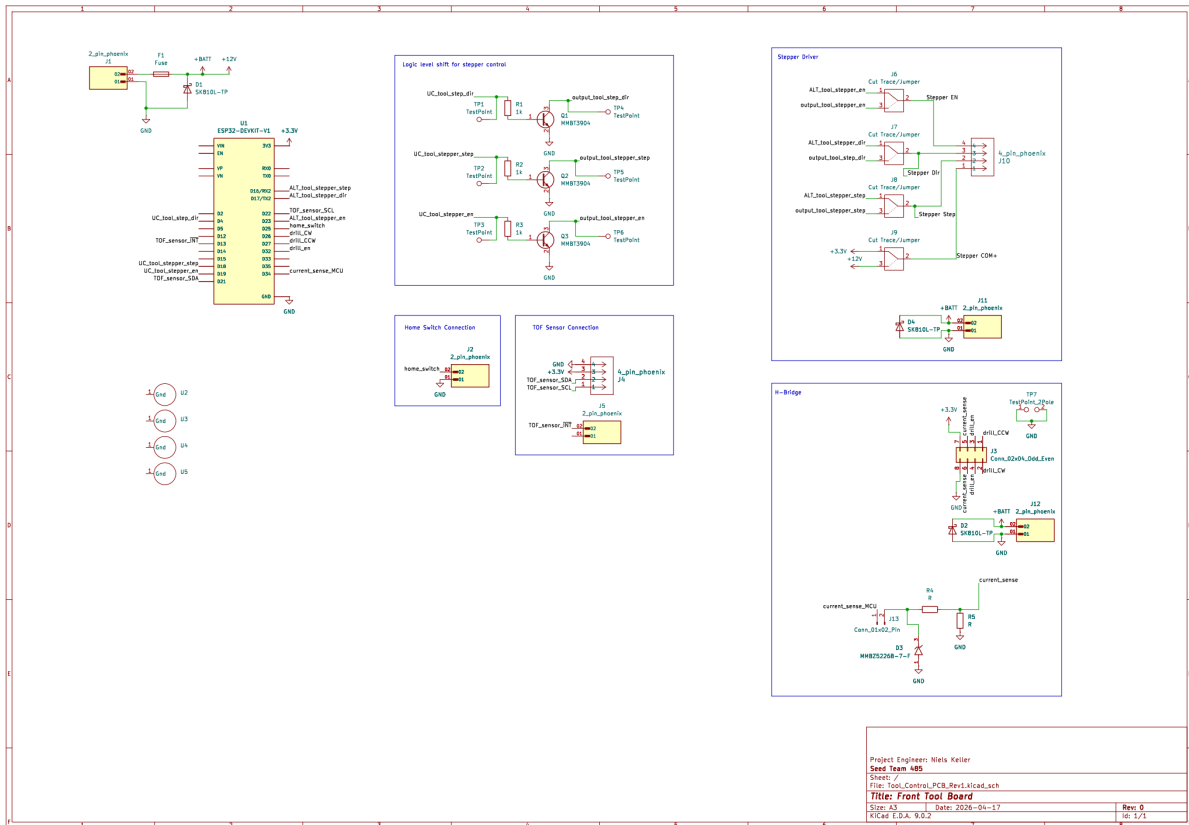


Figure 15. PCB Schematic

2.3.2 PCB Drawings

File containing the PCB schematic drawings.

Electrical\FrontTool\Tool_Control_PCB_Rev0_Layout.pdf

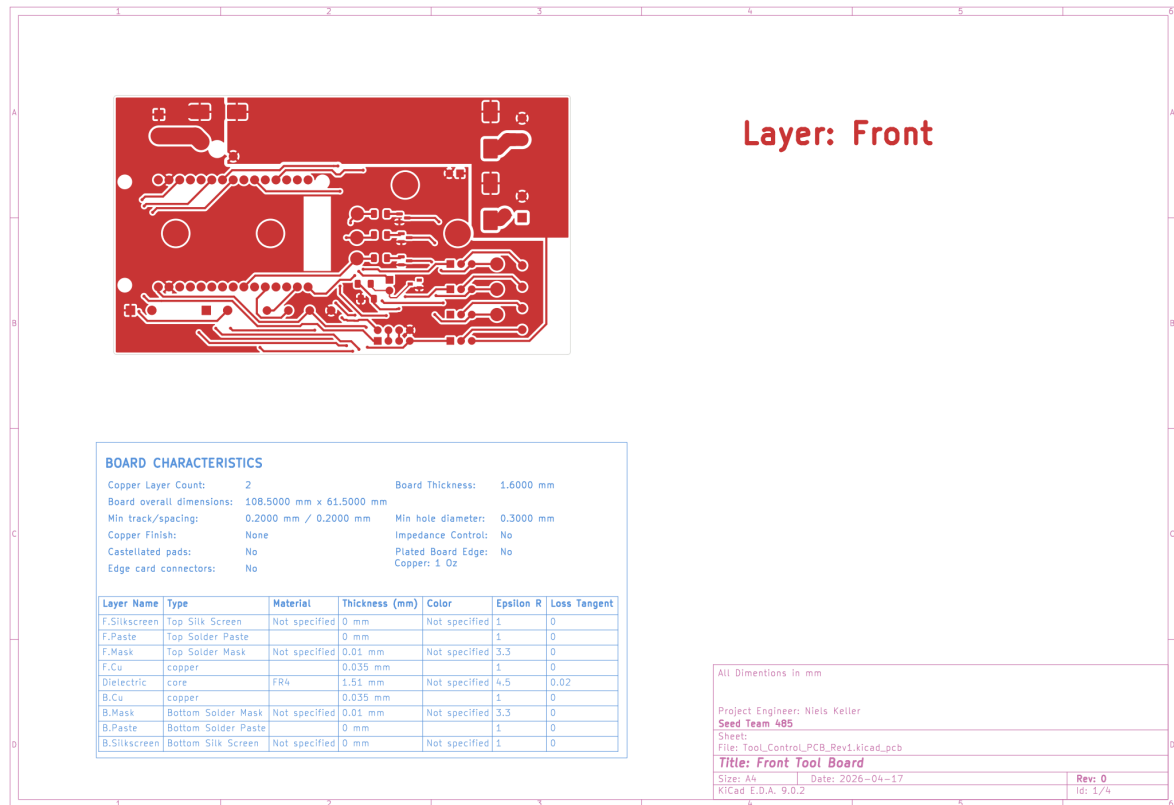


Figure 16. PCB Layout

2.3.3 PCB Bill of Materials

Files containing the bill of materials for the PCB. The BoM is also shown in **Table 3**.

CSV: Electrical\FrontTool\FrontToolBOM.xlsx

Interactive: Electrical\FrontTool\ibom.html

Table 3. Front Tool PCB Bill of Materials

Item	MFG PN	Qty	Description	Designator	MFG
<i>Connectors</i>					
1	2-pin Phoenix	5	2-Pin Phoenix Con- nector	J1, J2, J5, J11, J12	Phoenix Contact
4	1757268	2	Term Block Hdr 4- pos 5.08 mm	J4, J10	Phoenix Contact
7	Conn_02x04_Pins	1	2×4 Pin Header	J3	Generic
8	Conn_01x02_Pin	1	1×2 Pin Header	J13	Generic
12	SPC02SVGN-RC	1	Shorting Jumper 0.100 in Gold	N/A	Sullins
<i>Semiconductors</i>					
2	MMBZ5226B-7-F	1	Zener Diode 3.3 V 350 mW SOT-23	D3	Diodes Inc.
3	SK810L-TP	3	Schottky Diode 100 V 8 A SMC	D1, D2, D4	MCC
6	MMBT3904	3	NPN Transistor 40 V 200 mA SOT-23	Q1, Q2, Q3	onsemi / Diodes Inc.
<i>Passives</i>					
5	RNCP1206FTD1K00	3	Res 1 kΩ 1% 1/2 W 1206	R1, R2, R3	Stackpole
10	DNP	2	Do Not Populate	R4, R5	–
<i>Compute</i>					
9	ESP32-DEVKIT-V1	1	ESP32 Dev Board (Wi-Fi/BT MCU)	U1	Espressif
<i>Power & Protection</i>					
11	01530008Z	1	Fuse Holder Blade 32 V 15 A PCB	F1	Littelfuse
13	0297005.WXNV	1	Fuse Automotive 5 A 32 VDC Blade	N/A	Littelfuse

2.3.4 KiCad Files

Files containing the KiCad files related to the front tool board.

Electrical\FrontTool\front_tool_kicad.zip

3 Software Documentation

3.1 Front Tool

1. File containing the tool software for use with a laptop for testing.

`Software\Front_Board_ESP32\stepper_terminal\stepper_terminal.ino`

2. File containing the tool software for use with the Jetson. `Software\Front_Board_ESP32\front_tool_firmware\front_tool_firmware.ino`

3.2 ROS2 Packages

Each ROS2 package contains several ROS2 node scripts in .py format, a launch file in .py format, a setup file in .py format, and various build files that are created automatically from running the package. To run the package, only the location of the ROS workspace is needed. In this case, the ROS workspace is "seed_controller_ws". The following file paths direct to the overall package folders within the ROS workspace, and the listed .py files are within the package subfolder (i.e. `arducam_node.py` in the `demo_night` package refers to the absolute path `Software\ros_packages\seed_controller_ws\src\demo_night\demo_night\arducam_node` within the documentation package folder). Due to the extension amount of software on this project, for instructions on how to operate these ROS2 packages and how each package functions, see the User Manual.

1. `Software\ros_packages\seed_controller_ws\src\demo_night`

- `arducam_node.py`
- `aruco_rehoming_node.py`
- `commander_node.py`
- `display_node.py`
- `gtg_controller_node.py`
- `gui_node.py`
- `optical_path_follower_node.py`
- `PID.py`
- `realsense_node.py`
- `sabertooth.py`
- `tool_controller_node.py`

This package also contains the following data files:

- `arducam_calibration_data.npz`
- `calibration_data.npz`
- `dead_reckoning_cal.npz`

In addition to these nodes and scripts, the `demo_night` package contains four standalone .py files within the `Software\ros_packages\seed_controller_ws\src\demo_night` directory:

- `arducam_color_sampler.py`
- `aruco_generator.py`
- `dead_reckoning_calibration.py`
- `realsense_color_sampler.py`

2. `Software\ros_packages\seed_controller_ws\src\ml_red_controller`

- `arducam_detector_node.py`
- `cam_display_node.py`
- `gtg_controller_node.py`
- `gtg_motor_node.py`
- `ml_red_detector_node.py`
- `PID.py`
- `sabertooth.py`

This package also contains the following data files:

- `arducam_calibration_data.npz`
- `calibration_data.npz`

In addition to these nodes and scripts, the `ml_red_controller` package contains two standalone .py files within the `Software\ros_packages\seed_controller_ws\src\ml_red-controller` directory:

- `arducam_calibration.py`
- `BEVcalibration.py`

3. `Software\ros_packages\seed_controller_ws\src\motor_tester`

- `motor_test_node.py`
- `PID.py`
- `sabertooth.py`

4. `Software\ros_packages\seed_controller_ws\src\opti_waypoint_nav`

- `dual_camera_node.py`
- `optitrack_test_node.py`
- `path_follower_node.py`
- `PID.py`
- `sabertooth.py`
- `telemetry_node.py`

- `virtual_twin_node.py`
- `waypoint_manager_node.py`

5. `Software\ros_packages\seed_controller_ws\src\simple_teleop`

- `PID.py`
- `sabertooth.py`
- `teleop_node.py`

3.3 Machine Learning

The following file paths direct to the overall model folders (i.e. `detr_run.py` in the DETR folder refers to the absolute path `Software\MachineLearning\DETR\detr_run.py` within the documentation package folder). Each of these scripts has a description of the function in each of the scripts, and running these scripts can be done through any standard python IDE.

1. `Software\MachineLearning\DETR`

- `dandelion_detr_dataset.py`
- `detr_eval_confusion.py`
- `detr_run.py`
- `split_coco_dandelions.py`
- `train_detr_dandelions.py`

This folder also contains the following data files:

- `detr-dandelion\config.json`
- `detr-dandelion\preprocessor_config.json`

2. `Software\MachineLearning\DETRred`

- `detr_eval_red.py`
- `red_detr_dataset.py`
- `split_coco_red.py`
- `train_detr_red.py`

This folder also contains the following data files:

- `detr_red_model\config.json`
- `detr_red_model\preprocessor_config.json`

3. `Software\MachineLearning\YOLO`

- `TestingAnnotationswithYOLO.py`
- `testYOLO_model.py`
- `trainYOLO_model.py`

This folder also contains the following data files:

- dandelions.yaml
- yolov8n.pt
- runs\dandelion_train_v1\weights\best.pt

4. Software\MachineLearning\YOLOred

- TestingAnnotationswithYOLO.py
- testYOLO_model.py
- trainYOLO_model.py
- YOLO_data_splitting.py

This folder also contains the following data files:

- red.yaml
- yolov8n.pt
- runs\red_train_7\weights\best.pt

4 Design Files

GitHub was used for the majority of design files, including all software. The GitHub Repository used for this project can be found here:

<https://github.com/NielsKeller/SEED25>

Location	File Name	Description
EduSourced/Technical Documentation Package	Electrical.zip	All drawings and designs related to PCBs, electrical connections, and electrical systems
EduSourced/Technical Documentation Package	Mechanical.zip	All drawings and designs related to physical assemblies, 3D prints, custom parts, and water jet cut files
EduSourced/Technical Documentation Package	Software.zip	All code relating to front tool control, machine learning, autonomous behavior, and ROS2 packages
EduSourced/Technical Documentation Package	User_Manual.pdf	Document describing how to use the robot as a consumer. Not intended for describing engineering processes or aspects of the product not intended for direct consumer interface
EduSourced/Technical Documentation Package	TDPI.pdf	This document

5 User Manual

File containing the user manual.

User_Manual.pdf